ADAPTING PRE-TRAINED DIFFUSION MODEL FOR BLIND IMAGE DENOISING VIA NOISE COMPENSATION AND TIMESTEP PREDICTION

*Kenan Zou, Qianjun Huang, Jiaqing Wang, Kai Zhang**

School of Intelligence Science and Technology, Nanjing University

ABSTRACT

Diffusion models have recently achieved remarkable success in image restoration, particularly in image super-resolution. However, blind image denoising remains much less explored, despite being a fundamental problem in low-level vision. In this work, we propose an adaptive noise and timestep diffusion model for blind image denoising, named ANTD, which augments a pre-trained diffusion backbone with an auxiliary adaptive noise and timestep branch, referred to as ANT. Specifically, given a noisy input image, the ANT branch jointly predicts a noise compensation map and an appropriate diffusion timestep. The diffusion model then performs one-step denoising using the compensated noisy image and the predicted timestep. By training the framework end-to-end with a fixed diffusion backbone, ANTD inherits the rich generative prior of the pre-trained diffusion model and adapts it to blind image denoising. Extensive experiments on synthetic benchmarks demonstrate that ANTD achieves higher PSNR/SSIM and substantially improves perceptual quality, while also generalizing well to real-world image denoising.

***Index Terms*—** Blind Denoising, Diffusion Model, Noise Compensation, Timestep Prediction

1. INTRODUCTION

Image denoising aims to recover a clean image from a noisy observation. As a fundamental task in low-level vision, it remains a challenging ill-posed problem. Traditional algorithms [1] rely on handcrafted priors and often struggle with complex real-world noise. In recent years, deep discriminative models [2, 3, 4, 5, 6, 7] have achieved dominant performance. However, their pixel-wise losses, such as L_1 and MSE, tend to produce over-smoothed results and lose high-frequency details. In contrast, generative adversarial networks (GANs) [8] can generate richer textures, but they often suffer from training instability and visual artifacts. Recently, Diffusion Models [9, 10, 11] have emerged as powerful image priors for low-level vision. Despite the fundamental importance of blind image denoising, the use of diffusion priors for this task remains much less explored.

*Corresponding author (kaizhang@nju.edu.cn).

Directly applying diffusion priors to blind image denoising faces two main challenges. First, there exists a noise distribution mismatch between complex real-world noise and the standard Gaussian noise used in diffusion pre-training, which limits the effectiveness of the prior. Second, mapping unknown noise levels to diffusion timesteps is non-trivial, while multi-step sampling introduces high computational cost and may accumulate errors. Consequently, these issues hinder the advancement of diffusion models in blind image denoising.

To address these challenges, we propose an adaptive noise and timestep diffusion model (ANTD), a novel end-to-end framework that adapts pre-trained diffusion priors to blind image denoising. Rather than fine-tuning a large diffusion backbone [10] or relying on costly iterative sampling [12], ANTD keeps the pre-trained diffusion backbone frozen and exploits its intermediate states as a set of Gaussian denoisers with different denoising strengths. To make blind noisy inputs compatible with this frozen diffusion prior, we introduce a lightweight adaptive noise and timestep (ANT) branch to predict a noise compensation map and an appropriate diffusion timestep. The compensation map reduces the distribution gap between the noisy input and the diffusion input space, while the timestep selects the corresponding denoising strength.

With the predicted noise compensation map and timestep, ANTD performs one-step denoising through the frozen pre-trained diffusion backbone. This design avoids backbone fine-tuning and iterative sampling, while achieving high-fidelity reconstruction with rich image details.

The main contributions of this work are as follows:

- We propose ANTD, a novel end-to-end framework that leverages diffusion priors for blind denoising without fine-tuning the diffusion backbone.
- We design a lightweight ANT branch that adaptively predicts noise compensation map and appropriate timestep to explicitly align complex noisy image with the diffusion space.
- ANTD achieves state-of-the-art performance on blind Gaussian denoising with one-step inference and demonstrates competitive generalization to real-world scenarios compared with existing discriminative methods.

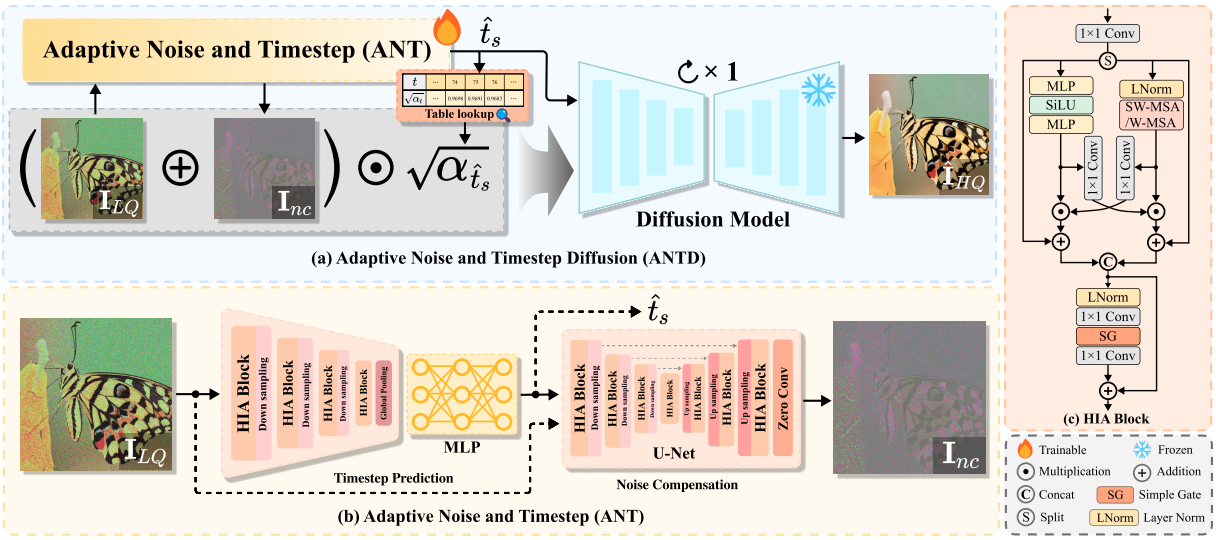


Fig. 1: The architecture of the proposed network. By adaptively predicting an appropriate timestep $\hat{t}_s$ and a noise compensation map I_{nc}, the ANT branch makes the distribution of the compensated noisy input consistent with the input space of the frozen pre-trained diffusion model.

2. RELATED WORK

2.1. Blind Image Denoising

Traditional denoising methods [1] have been extensively studied, but they often rely on handcrafted priors and struggle with complex noise. Recent discriminative methods, including CNN-based [2, 13, 3] and Transformer-based models [5, 6, 7], have achieved strong performance in blind denoising. However, these methods are usually trained with pixel-wise reconstruction losses, which tend to favor high fidelity scores but produce over-smoothed results.

2.2. Generative Models for Restoration

To improve perceptual quality, GANs [8] have been employed for image restoration, but they often suffer from training instability and visual artifacts. Recently, diffusion models [9, 10, 12] have demonstrated strong generative capabilities. However, applying diffusion models to blind denoising remains challenging because of the noise domain gap between real-world inputs and the pre-training distribution. In addition, iterative multi-step sampling is computationally expensive and tends to accumulate errors.

2.3. Efficient Utilization of Diffusion Priors

Although diffusion priors have been extensively applied to image restoration [9, 10, 12, 11], their potential for blind image denoising remains much less explored. Fine-tuning strategies [10, 11] incur high training costs, while iterative sampling methods [12] lead to heavy inference overhead. Methods such as DMID [14] use diffusion priors but require known noise levels.

3. METHODOLOGY

Inspired by [13] and [14], we view a pre-trained diffusion model as a Gaussian denoiser, where different intermediate states provide different denoising strengths. Based on this insight, we propose an adaptive noise and timestep diffusion model (ANTD), an end-to-end framework that adapts pre-trained diffusion priors to blind denoising. ANTD contains two key components: noise timestep prediction and noise compensation, as detailed below.

3.1. Noise Timestep Prediction

We treat pre-trained diffusion model as a Gaussian denoiser. The denoising level of its intermediate states depends on the timestep t. To leverage the diffusion prior, it is crucial to align the input image noise level σ with the timestep t.

For brevity, we first consider AWGN denoising. Given a noisy image $\mathbf{I}_{LQ}$ and a clean image $\mathbf{I}_{HQ}$, we have $\mathbf{I}_{LQ} = \mathbf{I}_{HQ} + \sigma \cdot \epsilon$, where $\epsilon \sim \mathcal{N}(0, 1)$. During the forward diffusion process, the intermediate state $\mathbf{x}_t$ is given by $\mathbf{x}_t = \sqrt{\alpha_t}\mathbf{x}_0 + \sqrt{1-\alpha_t}\epsilon$, where $\sqrt{\alpha_t}$ represents the cumulative noise schedule factor and $\mathbf{x}_0$ denotes the clean image. We scale the intermediate state $\mathbf{x}_t$ and observe that the degraded image $\mathbf{I}_{LQ}$ and $\mathbf{x}_t/\sqrt{\alpha_t}$ share the same mathematical form:

$$\mathbf{I}_{LQ} = \mathbf{I}_{HQ} + \sigma \cdot \epsilon, \quad \frac{\mathbf{x}_t}{\sqrt{\alpha_t}} = \mathbf{x}_0 + \frac{\sqrt{1-\alpha_t}}{\sqrt{\alpha_t}}\epsilon. \quad (1)$$

Essentially, denoising an image with noise level σ is equivalent to processing the intermediate state at timestep t, where $\sigma = \sqrt{1-\alpha_t}/\sqrt{\alpha_t}$. This observation establishes a correspondence between the diffusion timesteps $t \in [0, T]$ and the image noise levels σ. We calculate the noise level

$\sqrt{1-\alpha_t}/\sqrt{\alpha_t}$ for each timestep t as signal noise ratio schedule. For an image with noise level σ_s, we search this table to identify a timestep t_s such that $\sqrt{1-\alpha_{t_s}}/\sqrt{\alpha_{t_s}}$ is closest to σ_s. The noisy image $\mathbf{I}_{LQ}$ is then multiplied by $\sqrt{\alpha_{t_s}}$ to convert it into the intermediate state x_s. We use x_s as the starting state for one-step inference of the diffusion model.

In blind Gaussian denoising, the latent noise level σ of $\mathbf{I}_{LQ}$ is unknown. Consequently, we cannot directly lookup the table to find the optimal t. To address this, we transform the complex image restoration problem into a simpler timestep regression task. The corresponding noise level σ_s is computed as the standard deviation of the residual between noisy and the clean images, which is then mapped to the ground truth timestep t_s. As shown in Fig. 1(b), we employ a simple encoder to adaptively predict the target start timestep t_s directly from the noisy input $\mathbf{I}_{LQ}$.

3.2. Noise Compensation

Standard pre-trained diffusion models are inherently tailored for zero mean Gaussian noise. However, real-world noise distributions are complex and often non-Gaussian. This discrepancy creates a significant domain gap, leading to the challenge of input inconsistency. Drawing inspiration from [15], we propose a noise compensation mechanism. Our goal is to learn a noise compensation map $\mathbf{I}_{nc}$ to modulate the real noisy image $\mathbf{I}_{LQ}$. The noise distribution of $\mathbf{I}_{LQ} + \mathbf{I}_{nc}$ is intended to be consistent with the pre-trained diffusion space.

To estimating this map, we also employ a simple network as shown in Fig. 1(b). Conditioned on the input $\mathbf{I}_{LQ}$ and the predicted timestep $\hat{t}_s$, this network predicts the appropriate noise compensation map $\mathbf{I}_{nc}$. We do not strictly constrain $\mathbf{I}_{LQ}+\mathbf{I}_{nc}$ to a zero mean distribution. The optimization of $\mathbf{I}_{nc}$ is implicitly driven by the final reconstruction loss back propagated through the frozen diffusion model. Interestingly, the estimated $\mathbf{I}_{nc}$ exhibits a distinct spatially correlated with the input noise patterns in $\mathbf{I}_{LQ}$. This suggests that $\mathbf{I}_{nc}$ implicitly learns to counteract the non-Gaussian components of the real-world noise. By doing so, it renders the compensated image distribution statistically indistinguishable from the standard Gaussian noise expected by the frozen diffusion model.

3.3. Adaptive Noise and Timestep Module Design

Fig. 1 illustrates the overall architecture of our adaptive noise and timestep diffusion (ANTD) framework. The input noisy image $\mathbf{I}_{LQ}$ is first processed by the proposed ANT branch, which consists of a timestep prediction branch and a noise compensation branch. Within ANT, a encoder followed by global pooling is employed to predict the timestep $\hat{t}_s$. Simultaneously, the predicted $\hat{t}_s$ is expanded and concatenated with $\mathbf{I}_{LQ}$, and passed through a U-Net style network to estimate the noise compensation map $\mathbf{I}_{nc}$. Notably, we employ a zero initialized convolution at the output of the compensation branch. This ensures the module initiates as a null operation, preventing random perturbations and stabilizing early training. The entire forward process can be formulated as:

$$\mathbf{I}_{nc}, \hat{t}_s = \mathcal{F}_{\text{ANT}}(\mathbf{I}_{LQ}), \tag{2}$$

$$\hat{\mathbf{I}}_{HQ} = \mathcal{F}_{\text{Diff}}(\sqrt{\alpha_{\hat{t}_s}} \cdot (\mathbf{I}_{LQ} + \mathbf{I}_{nc}), \hat{t}_s), \tag{3}$$

where $\mathcal{F}_{\text{ANT}}$ denotes the trainable ANT network, and $\mathcal{F}_{\text{Diff}}$ represents the frozen pre-trained diffusion backbone. The aligned input $\sqrt{\alpha_{\hat{t}_s}} \cdot (\mathbf{I}_{LQ} + \mathbf{I}_{nc})$ is fed into $\mathcal{F}_{\text{Diff}}$ conditioning on $\hat{t}_s$ to reconstruct the clean image $\hat{\mathbf{I}}_{HQ}$. $\sqrt{\alpha_{\hat{t}_s}}$ is obtained by interpolating between the two adjacent timesteps.

As shown in Fig. 1, the ANT branch is built upon the hybrid interaction attention (HIA) block. HIA block adopts a parallel dual-branch architecture. An MLP branch targets pixel-independent noise via local features, while a Transformer branch utilizing SW-MSA [16] captures long-range dependencies for spatially correlated patterns. These complementary features are dynamically calibrated and fused via FFN and Simple Gate [17] for diverse noise patterns.

3.4. Loss Function

The ANT branch is trained end-to-end with a reconstruction loss and a timestep regression loss. The reconstruction loss is applied to the final denoised output $\hat{\mathbf{I}}_{HQ}$, and its gradients are back-propagated through the frozen diffusion backbone to optimize the generation of the compensation map $\mathbf{I}_{nc}$. For timestep regression, we minimize the L_2 distance between the predicted timestep $\hat{t}_s$ and the ground-truth timestep t_s, where t_s is obtained by mapping the standard deviation of the residual between the noisy input and the ground-truth clean image to the diffusion timestep schedule. The total loss is defined as:

$$\mathcal{L}_{total} = \underbrace{\|\hat{\mathbf{I}}_{HQ} - \mathbf{I}_{HQ}\|^2}_{\mathcal{L}_{rec}} + \lambda \cdot \underbrace{\|\hat{t}_s - t_s\|^2}_{\mathcal{L}_t}, \tag{4}$$

where λ balances timestep regression and image reconstruction, and is empirically set to 1.0.

4. EXPERIMENTS

We use the open-source pre-trained unconditional diffusion model from [19], which is trained on ImageNet at a resolution of 256×256 with $T = 1000$ timesteps. For training, we use a combined dataset consisting of Exploration [20], DIV2K [21], and Flickr2K [22]. The diffusion backbone, which contains 552.8M parameters, is kept frozen, and only the lightweight ANT branch with 5.6M parameters is optimized. During training, denoised images are recovered by one-step sampling. We use the Adam optimizer with a patch size of 256×256 and a batch size of 4 on 2 NVIDIA RTX 4090 GPUs.

Table 1: Quantitative comparison of blind Gaussian image denoising on four benchmark datasets. PSNR (dB)↑, SSIM↑, and LPIPS↓ are reported. The best and second-best results are highlighted in red and blue, respectively.

Dataset	Noise Level	DRUNet [3]	Restormer [6]	Xformer [7]	CLIPDenoising [18]	**ANTD (Ours)**
		PSNR / SSIM / LPIPS	PSNR / SSIM / LPIPS	PSNR / SSIM / LPIPS	PSNR / SSIM / LPIPS	PSNR / SSIM / LPIPS
CBSD68	$\sigma = 15$	34.30 / 0.9344 / 0.0555	34.39 / 0.9354 / 0.0544	34.41 / 0.9357 / 0.0536	33.97 / 0.9303 / 0.0571	34.46 / 0.9362 / 0.0500
	$\sigma = 25$	31.69 / 0.8926 / 0.0962	31.78 / 0.8941 / 0.0939	31.80 / 0.8945 / 0.0936	31.02 / 0.8775 / 0.1063	31.86 / 0.8958 / 0.0839
	$\sigma = 50$	28.51 / 0.8103 / 0.1815	28.59 / 0.8131 / 0.1771	28.60 / 0.8135 / 0.1766	26.69 / 0.7307 / 0.2816	28.71 / 0.8164 / 0.1622
	$\sigma = 100$	25.75 / 0.7061 / 0.3037	25.84 / 0.7113 / 0.2895	25.81 / 0.7107 / 0.2939	22.82 / 0.5249 / 0.4975	25.97 / 0.7161 / 0.2677
	$\sigma = 150$	24.30 / 0.6428 / 0.3864	24.42 / 0.6496 / 0.3647	24.35 / 0.6482 / 0.3708	20.18 / 0.3830 / 0.6134	24.53 / 0.6541 / 0.3436
	$\sigma = 200$	23.34 / 0.5997 / 0.4494	23.47 / 0.6075 / 0.4230	23.38 / 0.6056 / 0.4268	18.15 / 0.2880 / 0.6811	23.57 / 0.6115 / 0.4018
Kodak24	$\sigma = 15$	35.31 / 0.9292 / 0.0710	35.44 / 0.9304 / 0.0717	35.48 / 0.9307 / 0.0715	34.69 / 0.9218 / 0.0798	35.50 / 0.9311 / 0.0627
	$\sigma = 25$	32.89 / 0.8917 / 0.1103	33.02 / 0.8936 / 0.1101	33.07 / 0.8940 / 0.1097	31.86 / 0.8710 / 0.1264	33.10 / 0.8949 / 0.0966
	$\sigma = 50$	29.87 / 0.8208 / 0.1870	30.00 / 0.8235 / 0.1851	30.09 / 0.8254 / 0.1840	27.39 / 0.7226 / 0.3351	30.12 / 0.8264 / 0.1698
	$\sigma = 100$	27.14 / 0.7341 / 0.2957	27.31 / 0.7391 / 0.2895	27.36 / 0.7405 / 0.2857	23.22 / 0.5085 / 0.5428	27.48 / 0.7438 / 0.2636
	$\sigma = 150$	25.65 / 0.6817 / 0.3764	25.86 / 0.6876 / 0.3605	25.88 / 0.6886 / 0.3620	20.35 / 0.3570 / 0.6520	26.05 / 0.6937 / 0.3329
	$\sigma = 200$	24.62 / 0.6471 / 0.4380	24.85 / 0.6538 / 0.4188	24.87 / 0.6535 / 0.4240	18.16 / 0.2584 / 0.7176	25.08 / 0.6600 / 0.3861
McMaster	$\sigma = 15$	35.41 / 0.9324 / 0.0513	35.55 / 0.9339 / 0.0521	35.67 / 0.9352 / 0.0499	33.86 / 0.9100 / 0.0654	35.72 / 0.9358 / 0.0464
	$\sigma = 25$	33.15 / 0.9030 / 0.0810	33.31 / 0.9057 / 0.0801	33.42 / 0.9077 / 0.0762	31.50 / 0.8660 / 0.0945	33.48 / 0.9080 / 0.0687
	$\sigma = 50$	30.08 / 0.8457 / 0.1401	30.29 / 0.8516 / 0.1344	30.36 / 0.8538 / 0.1317	27.43 / 0.7272 / 0.2311	30.48 / 0.8551 / 0.1256
	$\sigma = 100$	27.05 / 0.7664 / 0.2310	27.29 / 0.7778 / 0.2177	27.32 / 0.7787 / 0.2146	23.45 / 0.5304 / 0.4202	27.54 / 0.7830 / 0.2005
	$\sigma = 150$	25.34 / 0.7119 / 0.2981	25.61 / 0.7280 / 0.2778	25.58 / 0.7265 / 0.2791	20.71 / 0.3871 / 0.5437	25.87 / 0.7333 / 0.2543
	$\sigma = 200$	24.17 / 0.6703 / 0.3502	24.45 / 0.6904 / 0.3258	24.40 / 0.6880 / 0.3284	18.63 / 0.2912 / 0.6235	24.72 / 0.6953 / 0.2992
Urban100	$\sigma = 15$	34.83 / 0.9505 / 0.0319	35.06 / 0.9519 / 0.0330	35.29 / 0.9535 / 0.0317	33.15 / 0.9296 / 0.0458	35.24 / 0.9531 / 0.0298
	$\sigma = 25$	32.61 / 0.9283 / 0.0516	32.91 / 0.9308 / 0.0515	33.21 / 0.9333 / 0.0492	30.72 / 0.8930 / 0.0716	33.09 / 0.9326 / 0.0455
	$\sigma = 50$	29.61 / 0.8835 / 0.0925	30.02 / 0.8894 / 0.0898	30.36 / 0.8940 / 0.0855	26.27 / 0.7693 / 0.2063	30.28 / 0.8935 / 0.0818
	$\sigma = 100$	26.50 / 0.8115 / 0.1619	27.04 / 0.8249 / 0.1514	27.36 / 0.8316 / 0.1448	22.15 / 0.5727 / 0.4161	27.41 / 0.8328 / 0.1381
	$\sigma = 150$	24.63 / 0.7526 / 0.2229	25.23 / 0.7718 / 0.2045	25.49 / 0.7784 / 0.1977	19.58 / 0.4309 / 0.5581	25.65 / 0.7822 / 0.1853
	$\sigma = 200$	23.31 / 0.7033 / 0.2762	23.93 / 0.7262 / 0.2517	24.12 / 0.7319 / 0.2461	17.67 / 0.3348 / 0.6508	24.39 / 0.7391 / 0.2258

4.1. Blind Gaussian Denoising

We first evaluate ANTD on blind Gaussian denoising. The ANT branch is trained with noise levels $\sigma \in [5, 250]$, which is much broader than the commonly used range of $\sigma \in [0, 50]$. Such a wide range is more challenging, as it can weaken performance at specific noise levels.

We conduct experiments on four widely used benchmarks: CBSD68 [23], Kodak24 [24], McMaster [25], and Urban100 [26]. ANTD is compared with representative blind Gaussian denoising methods, including DRUNet [3], Restormer [6], Xformer [7], and CLIPDenoising [18], using PSNR (dB), SSIM, and LPIPS [27]. For fair comparison, we rerun all methods with their official pre-trained weights under the same evaluation setting.

Since the public models of these methods are mainly trained for noise levels up to $\sigma = 50$, we follow the strategy of DRUNet [3] during testing and rescale higher noise levels to have a standard deviation of 50. As shown in Table 1 and Fig. 2, ANTD achieves state-of-the-art performance on most datasets and remains robust under severe noise. In particular, the superior LPIPS scores indicate that the diffusion prior effectively helps recover rich image details. Although CLIPDenoising [18] uses CLIP encoder priors, its performance degrades significantly at high noise levels. In contrast, ANTD maintains strong robustness and delivers better results in these challenging cases.

Table 2: Comparison of real-world image denoising on the SIDD dataset.

Method	PSNR	SSIM	LPIPS	MANIQA	CLIPIQA	MUSIQ
MIRNet [28]	39.43	0.9189	0.2018	0.2907	0.2843	22.06
MAXIM [4]	39.68	0.9217	0.1894	0.2973	0.2914	22.39
Uformer [5]	39.81	0.9205	0.2001	0.2971	0.2940	22.39
Restormer [6]	39.93	0.9215	0.1976	0.2989	0.2920	22.52
Xformer [7]	39.89	0.9215	0.1974	0.2985	0.2894	22.54
CLIPDenoising [18]	34.79	0.8662	0.2349	0.2319	0.2814	24.40
GCP-ID [1]	35.29	0.8953	0.2392	0.2375	0.2758	19.76
ANTD (Ours)	39.37	0.9164	0.2145	0.2951	0.2955	22.55

4.2. Real-World Denoising

To evaluate the generalization ability of ANTD, we train it on SIDD-Medium [29] and compare it with representative discriminative models, including MIRNet [28], MAXIM [4], Uformer [5], Restormer [6], and Xformer [7], as well as the traditional method GCP-ID [1] and CLIPDenoising [18]. For a fair comparison, all methods are evaluated on the SIDD validation PNG images using their official weights.

As shown in Table 2, ANTD achieves slightly lower full-reference scores, including PSNR, SSIM, and LPIPS, than discriminative models, but obtains promising results on no-reference perceptual metrics, including MANIQA [30], CLIPIQA [31], and MUSIQ [32]. Most discriminative methods train the entire network specifically for denoising, whereas ANTD uses a frozen diffusion backbone trained on Gaussian denoising and only learns the ANT branch to

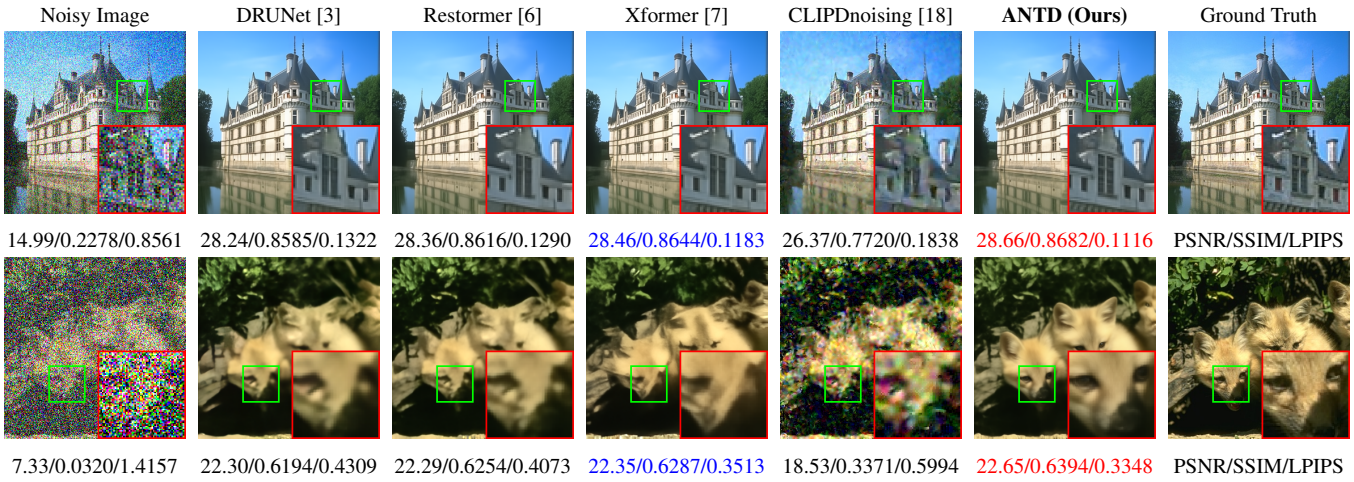


Fig. 2: Quantitative and visual comparisons for blind Gaussian denoising. Top row: $\sigma = 50$; Bottom row: $\sigma = 200$. PSNR (dB)↑, SSIM↑, and LPIPS↓ are provided. Leveraging diffusion priors, ANTD generates denoised images with richer details.

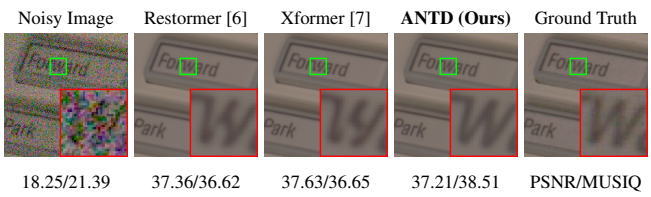


Fig. 3: Visual comparisons for real-world image denoising. ANTD achieves high perceptual quality while maintaining low distortion. PSNR (dB) and MUSIQ metrics are provided.

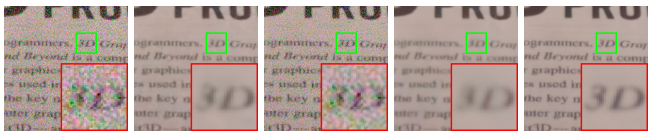


Fig. 4: Visual comparisons for the ablation study of the ANT branch with different components.

align the input state with the diffusion space. Despite this lightweight adaptation, ANTD achieves competitive performance against state-of-the-art baselines. Visual comparisons in Fig. 3 further show that ANTD produces visually pleasing results, with a good balance between detail recovery and image fidelity.

4.3. Ablation Study

We conduct ablation studies to evaluate the impact of the key components of ANTD, including timestep prediction, noise compensation, and the zero initialized convolution layer. Experiments are performed on synthetic CBSD68 with $\sigma = 50$ and real-world SIDD. When removing the timestep prediction module, we fix the embedding timestep to the average timestep of the training set.

As shown in Table 3 and Fig. 4, timestep prediction alone achieves reasonable performance on synthetic noise but is less effective on real-world noise. In contrast, using only noise compensation cannot establish a reliable correspondence between the noisy input and the pre-trained diffusion model. The zero initialized convolution layer further improves the accuracy and stability of noise compensation.

Table 3: Ablation study of designed ANT branch with different components. Comparisons of performance across both synthetic CBSD68 and real-world SIDD datasets.

Timestep Prediction	Noise Compensation	Zero Conv	CBSD68 PSNR / SSIM / LPIPS	SIDD PSNR / SSIM / LPIPS
✗	✓	✓	28.28 / 0.8002 / 0.1941	38.80 / 0.9085 / 0.2627
✓	✗	✗	28.71 / 0.8151 / 0.1695	27.98 / 0.5442 / 0.5016
✓	✓	✗	28.70 / 0.8149 / 0.1749	35.88 / 0.8625 / 0.2358
✓	✓	✓	28.71 / 0.8164 / 0.1622	39.37 / 0.9164 / 0.2145

5. CONCLUSION AND LIMITATIONS

In this paper, we proposed ANTD, an end-to-end framework that adapts frozen diffusion priors to blind image denoising. The proposed ANT branch predicts adaptive timesteps and noise compensation maps to align complex noisy inputs with the diffusion input space, enabling the frozen backbone to produce high-quality denoising results with one-step inference. Experiments showed that ANTD achieved state-of-the-art performance on blind Gaussian denoising and demonstrated promising generalization to real-world noise. Future work could focus on improving the efficiency and robustness of ANTD. We hope this work encourages further exploration of pre-trained generative priors for image restoration.

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